

Problem 14

Evaluate the integral:

$$\int \frac{2^{x+1} - 5^{x-1}}{10^x} dx.$$

Solution

To solve the given integral $\int \frac{2^{x+1} - 5^{x-1}}{10^x} dx$, we begin by simplifying the terms in the integrand.

Step 1: Simplify the terms

The integrand is:

$$\frac{2^{x+1} - 5^{x-1}}{10^x}.$$

Using the property of exponents $a^{b+c} = a^b \cdot a^c$, rewrite 2^{x+1} and 5^{x-1} :

$$2^{x+1} = 2^x \cdot 2^1 = 2 \cdot 2^x, \quad 5^{x-1} = 5^x \cdot 5^{-1} = \frac{5^x}{5}.$$

Substitute these into the integrand:

$$\frac{2^{x+1} - 5^{x-1}}{10^x} = \frac{2 \cdot 2^x}{10^x} - \frac{\frac{5^x}{5}}{10^x}.$$

Using the property of exponents $\frac{a^m}{a^n} = a^{m-n}$, simplify each term:

$$\frac{2 \cdot 2^x}{10^x} = 2 \cdot \frac{2^x}{10^x} = 2 \cdot \left(\frac{2}{10}\right)^x = 2 \cdot \left(\frac{1}{5}\right)^x,$$

$$\frac{\frac{5^x}{5}}{10^x} = \frac{5^x}{5 \cdot 10^x} = \frac{1}{5} \cdot \frac{5^x}{10^x} = \frac{1}{5} \cdot \left(\frac{5}{10}\right)^x = \frac{1}{5} \cdot \left(\frac{1}{2}\right)^x.$$

Thus, the integrand becomes:

$$\frac{2^{x+1} - 5^{x-1}}{10^x} = 2 \cdot \left(\frac{1}{5}\right)^x - \frac{1}{5} \cdot \left(\frac{1}{2}\right)^x.$$

Step 2: Rewrite the integral

The integral can now be written as:

$$\int \frac{2^{x+1} - 5^{x-1}}{10^x} dx = \int 2 \cdot \left(\frac{1}{5}\right)^x dx - \int \frac{1}{5} \cdot \left(\frac{1}{2}\right)^x dx.$$

Step 3: Solve each term

We evaluate each integral separately using the formula for the integral of an exponential function:

$$\int a^x dx = \frac{a^x}{\ln(a)} + C, \quad \text{for } a > 0, a \neq 1.$$

1. **First term: $\int 2 \cdot \left(\frac{1}{5}\right)^x dx$:** Let $a = \frac{1}{5}$. Using the constant multiple rule:

$$\int 2 \cdot \left(\frac{1}{5}\right)^x dx = 2 \cdot \int \left(\frac{1}{5}\right)^x dx = 2 \cdot \frac{\left(\frac{1}{5}\right)^x}{\ln\left(\frac{1}{5}\right)}.$$

Simplify $\ln\left(\frac{1}{5}\right) = -\ln(5)$:

$$\int 2 \cdot \left(\frac{1}{5}\right)^x dx = -\frac{2 \cdot \left(\frac{1}{5}\right)^x}{\ln(5)}.$$

2. **Second term: $\int \frac{1}{5} \cdot \left(\frac{1}{2}\right)^x dx$:** Let $a = \frac{1}{2}$. Using the constant multiple rule:

$$\int \frac{1}{5} \cdot \left(\frac{1}{2}\right)^x dx = \frac{1}{5} \cdot \int \left(\frac{1}{2}\right)^x dx = \frac{1}{5} \cdot \frac{\left(\frac{1}{2}\right)^x}{\ln\left(\frac{1}{2}\right)}.$$

Simplify $\ln\left(\frac{1}{2}\right) = -\ln(2)$:

$$\int \frac{1}{5} \cdot \left(\frac{1}{2}\right)^x dx = -\frac{\left(\frac{1}{2}\right)^x}{5 \cdot \ln(2)}.$$

Step 4: Combine the results

Adding the two results, the solution to the integral is:

$$\int \frac{2^{x+1} - 5^{x-1}}{10^x} dx = -\frac{2 \cdot \left(\frac{1}{5}\right)^x}{\ln(5)} - \frac{\left(\frac{1}{2}\right)^x}{5 \cdot \ln(2)} + C,$$

where C is the constant of integration.

Final Answer

The solution to the integral is:

$$\int \frac{2^{x+1} - 5^{x-1}}{10^x} dx = -\frac{2 \cdot \left(\frac{1}{5}\right)^x}{\ln(5)} - \frac{\left(\frac{1}{2}\right)^x}{5 \cdot \ln(2)} + C.$$